AIMLCZG512 - Deep Reinforcement Learning

Comprehensive Exam Syllabus and Study Guidelines

(1) Text book Reading Guide

Module	What to read from TB Chapters
Introduction: Introducing RL	Chapter #1,
	Chapter #2 (Sections 2.1 to 2.7)
Finite MDP + Dynamic Programming	Chapter #3 (Sections 3.1,3.2, 3.3, 3.5, 3.6)
	Chapter #4 (Sections 4.1,4.2,4.3) Chapter #4 (Sections 4.4,4.5,4.6)
Monte Carlo Methods	Chapter #5 (Sections 5.1 to 5.7) – Use the slides provided to plan the depth of topics
Temporal Difference Learning	Chapter #6 (Sections 6.1 to 6.7) - Use the slides provided to plan the depth of topics
	Chapter #7 (Section 7.1) – This was covered along with the later classes
On-policy Prediction with Approximation	Chapter #9 (Sections 9.1 to 9.7) – Use the slides provided to plan the depth of topics
Policy Gradient Methods	Chapter #13 (Sections 13.1 to 13.5) – Use the slides provided to plan the depth of topics
Special Topics	DQN – Use the directions from course page
	DDQN - Use the directions from course page
	MCTS – (Section 8.10, 8.11) Use Slides as guide
	UCB Action Section – Slides
	AlphaGo, AlphaGo Zero - Use the directions from course page
	Imitation Learning – Common Class – Used slides as guide

(2) Question Paper Pattern

- ✔ 40% Weightage
- ✔ 6 Questions (Each question be of 5/6/7 % Weightage) such that total is 40%
- ✔ All the topics listed as in the syllabus for exam (about 50% emphasis from pre-mid-term and 50% emphasis on post mid-term topics)
- ✔ Questions will be conceptual, problem-solving and applications based.
- ✔ Use the slide as guidance to read.
- ✔ Numerical and problem solving will test your ability to apply the algorithms to scenarios

(3) Sample Question Papers

We are providing past comprehensive exam question papers for reference. Please consider them as samples only, as they do not necessarily reflect the questions that may appear in future exams.

(4) **Finally**

Exams are just three weeks away! Start preparing now by focusing on understanding the topics, not just solving past papers. For any doubts, contact your instructors via Teams or email (not the day before exams).

Best wishes for your exams!

– DRL Team